FENSe: Feedback-Enabled Neighbor Selection for Spatial Aware Collaborative Perception

Qianxun Xu
Duke Kunshan University
qianxun.xu@dukekunshan.edu.cn

Azzedine Boukerche
University of Ottawa
boukerch@site.uottawa.ca

Peng Sun*
Duke Kunshan University
peng.sun568@dukekunshan.edu.cn

***Abstract*—Collaborative perception aims to enhance the ego vehicle's detection capability by integrating valuable perception data from neighbor vehicles. A critical challenge in collaborative perception systems is managing the trade-off between communication overhead and detection accuracy. To address this issue, traditional methods optimize feature selection typically by selectively transmitting high-confidence features from neighboring vehicles to complement the ego vehicle. However, these mechanisms often overlook redundancy and inconsistency among shared data, resulting in increased communication costs and reduced overall efficiency and accuracy. To overcome this limitation, we propose FENSe (Feedback-Enabled Neighbor Selection), a lightweight module that turns single-vehicle perception into a feedback signal for who should communicate, and thus inform the final collaborative perception results. The novelty of FENSe lies in three aspects: (1) It performs density-aware clustering of neighboring vehicles using spatial proximity and scene context to group agents with overlapping fields of view. (2) It selects one representative per cluster by estimating each neighbor's informativeness from single-vehicle detections, with decoupled criteria during training and inference. (3) It is plug-and-play with standard intermediate-fusion pipelines, requiring minimal modification. Extensive experiments demonstrate that FENSe substantially improves detection accuracy while decreasing communication volume and computation time required at the ego vehicle.**

***Index Terms*—Internet-of-Vehicles, autonomous driving, vehicular edge intelligence, collaborative perception, clustering**

I. Introduction

In autonomous driving, ensuring accurate and timely environmental perception is critical for safe navigation. However, traditional single-agent perception systems [1], [2] encounter inherent limitations, including restricted sensing range, occlusions, and performance degradation under adverse conditions, which can obscure critical information about the vehicle's surroundings. Recently, multi-agent collaborative perception has emerged as an effective strategy to address the shortcomings of individual perception systems. Leveraging advancements in Vehicle-to-Everything (V2X) communication and intelligent transportation systems, collaborative perception enables neighbor vehicles to share complementary perceptual information to the ego vehicle, which integrates the information and achieves a more comprehensive understanding of the driving environment.

Despite these advancements, collaborative perception faces critical challenges, primarily centered around optimizing the trade-off between communication efficiency and perception accuracy. Real-world communication networks impose stringent bandwidth constraints, so that it is impractical to transfer extensive raw sensor data or voluminous feature representations continuously. Consequently, many existing methods adopt selective communication strategies, such as handshake mechanisms, where the ego vehicle requests information, and neighboring vehicles respond based on perceived complementarities [3]–[5]. However, these approaches often neglect the inherent redundancy and inconsistency in data shared by multiple neighboring vehicles, leading to unnecessary communication overhead and suboptimal perception performance. Additionally, the request-process-respond mechanism during the handshake process significantly increases the communication time required. Current solutions addressing redundancy typically rely on methods such as prioritization of individual agents via calculating the priority weights of the overlap between the neighbors and the ego vehicle [6] or feature decoupling [7]. Yet, these methods are usually computationally intensive and insufficiently leverage spatial correlations among neighboring vehicles.

To overcome these limitations and utilize the spatial relationships, we introduce FENSe, a lightweight Feedback-Enabled dynamic Neighbor Selection module, to improve collaborative perception efficiency. Briefly, our method performs dynamic clustering of neighboring vehicles based on their spatial proximity and scene characteristics to group vehicles with similar viewpoints. The key insight recognizes that vehicles within spatial clusters share highly correlated observation perspectives, so selecting one representative per cluster avoids redundant transmissions while preserving informative cues. Once clusters are formed, FENSe evaluates the informativeness of each neighbor based on their single-vehicle detection results. The selection strategy is decoupled between training and inference for better adaptability. Finally, only the most informative vehicle per cluster is retained for feature fusion.

Furthermore, our dynamic neighbor selection module can be seamlessly integrated into various collaborative perception architectures as a practical and efficient plug-and-play component. In this work, we demonstrate the performance of FENSe by integrating it to a SCOPE [8] framework, and evaluate on the OPV2V dataset [9]. Experiment results show that this design of filtering redundant or less informative vehicles out-

* Corresponding author.

performs the raw [8] model by improving perception accuracy and simultaneously decreasing computation time at the ego vehicle and communication volume required.

Our main contributions can be summarized as follows:

1) We propose a spatially aware and density adaptive clustering scheme, grouping nearby vehicles that share overlapping viewpoints using spatial proximity and scene density. This balances scene coverage with cross neighbor overlap and provides structured inputs for the downstream selection stage.
2) We propose a decoupled representative selection policy. During training the representative in each cluster is the vehicle that contains the largest number of ground truth bounding boxes. During inference it is the vehicle with the highest aggregated confidence. Only these representatives transmit intermediate features to the ego.
3) Our module is lightweight and integrates as a plug and play component into collaborative perception pipelines that support intermediate feature fusion. On OPV2V [9] with a SCOPE [8] backbone it delivers a stronger accuracy and efficiency trade off while reducing ego side computation time and communication volume.

II. Related Work

Collaborative perception has emerged as a pivotal paradigm to overcome the inherent limitations of single-agent systems, particularly in occluded or sensor-limited environments. Early fusion [10]–[12] methods share raw sensor data across vehicles, which can lead to very high detection accuracy by providing a more comprehensive view of the environment to the ego vehicle. However, this comes at the cost of considerable communication overhead and often imposes strict bandwidth constraints. In contrast, late fusion [13]–[15] techniques transmit only the final detection outputs from each vehicle, thereby drastically reducing the amount of data exchanged. However, once local detection errors occur, they cannot be corrected by the ego vehicle, and missing cues from occluded objects are irretrievable. Intermediate fusion [3]–[9], [16]–[25] that transmit processed feature representations has gained more attention due to its optimization of the balance between accuracy and transmission bandwidth.

A wide range of methods have been proposed to enhance intermediate fusion. For example, F-Cooper [16] applies element-wise feature aggregation to combine spatially aligned representations. In V2VNet [17], a graph neural network is introduced to perform multi-round message passing, which enables each agent to iteratively refine its feature map from the information of all the nearby vehicles. DiscoNet [18] utilizes a teacher–student framework that distills rich representations from both early and intermediate fusion. In V2X-ViT [19], a heterogeneous multi-agent attention module is used to account for differences in sensor modalities across connected agents. Building on these foundations, recent transformer-based intermediate fusion methods emphasize cross-agent BEV reasoning, hetero-modal alignment, and instance or query level fusion. CoBEVT [20] designs a fused-axial attention to capture cross-agent spatial interactions for BEV semantics and 3D detection, improving multi-agent BEV fusion efficiency. HM-ViT [21] extends to hetero-modal agents with heterogeneous attention layers for robust V2V fusion, and TransIFF [22] performs instance-level transformer fusion for V2I with cross-domain adaptation to reduce domain gaps. While intermediate fusion reduces data volume relative to raw sharing, its cost remains significant, and redundant features are often exchanged between vehicles with overlapping viewpoints.

Beyond improving fusion architectures, recent work has increasingly emphasized selective communication to mitigate bandwidth and computation burdens. Methods such as When2com [3] and Who2com [4] propose handshake-style protocols, where the ego vehicle requests data and neighboring vehicles selectively respond based on perceived relevance. Similarly, Where2comm [5] employs a spatial confidence aware communication module for sharing only high-confidence information that the ego needs. How2comm [23] advances this line by maximizing informative content via mutual-information–aware spatial channel filtering, improving robustness under bandwidth limits. CoSDH [24] further refines where and what to communicate via supply demand awareness with an intermediate–late hybrid scheme, improving the accuracy bandwidth trade-off. SlimComm [25] leverages radar guided sparse queries to trigger communication on Doppler-salient regions, which also serves as a powerful where and what signal. While effective in limiting unnecessary transmissions between ego vehicle and its neighbors, these schemes focus more on the message content quality and often fail to consider the spatial correlation between multiple neighboring vehicles. This limitation leads to redundant transmissions from vehicles with overlapping views, especially in densely populated environments. Some solutions attempt to address this by computing priority weights based on overlap with the ego vehicle's field of view [6] or by performing feature decoupling to isolate novel information [7]. However, these strategies can be computationally expensive and involve additional processing delays due to multi-stage request–response communication protocols.

The tension between communication overhead and detection accuracy is equally central in the V2X-enabled collaborative perception, which addresses it through standards for interoperable perception exchange, utility aware sender selection with admission and scheduling, and systems that account for delay and loss. ETSI [30] defines interoperable object level messages and prescribes generation rules that adapt content and emission rate to channel conditions, which creates a common application layer for vehicles and roadside units to exchange machine readable perceptions with predictable load. On this foundation, selection and timing become first class control surfaces. Online scheduling for cooperative perception chooses transmitters that add complementary views under time varying channels [31], and edge assisted coordination demonstrates that infrastructure guided orchestration can raise throughput and efficiency over fully distributed access when perception flows contend for scarce airtime [32].

Recent edge computing studies sharpen this view by making the cost of acquiring state, the freshness of information, and the decision to offload explicit optimization variables. In [33], the authors shows that status probing itself consumes resources and that probing frequency and offloading gains must be co optimized to achieve the best end to end efficiency at the network edge. In parallel, the authors in [34] formulates a joint problem that couples detection accuracy with information freshness and offloading choice, revealing that stale updates can negate the benefit of additional computation even when bandwidth is available. Vision oriented pipelines arrive at a compatible conclusion. Latency aware designs align features across time to avoid stale fusion under nonzero delays [35], and interruption aware methods reconstruct missing cooperative information when packets are lost, which improves robustness on public benchmarks [36].

In this paper, we propose FENSe, a module that leverages the correlation between spatially approximate neighbor vehicles to filter information on the vehicle level, effectively minimizing communication volume and computation costs while increasing detection accuracy.

III. Methodology

In collaborative perception systems, effectively identifying the most informative neighboring vehicles is crucial for balancing communication efficiency and perceptual accuracy. To address this, FENSe leverages the spatial locations of neighboring vehicles and scene context to dynamically cluster and select representative collaborators. This section is composed of an introduction of overall architecture, followed by the cluster and selection mechanism, and the detailed implementations at training and inference time.

A. Overall Architecture

Our framework is built upon the principle that spatially proximate vehicles often share overlapping fields of view and thus provide correlated perceptual information. Consequently, selecting a representative from each spatial region can sufficiently convey the environmental information of that region, and collectively, these representatives form a complete perception of the entire scene. Formally, we denote an ego vehicle and N neighboring vehicles in a scene as $\mathcal{V} = \{v_e\} \cup \{v_n\}_{n=1}^{N}$, each with positions $\mathbf{p}_i \in \mathbb{R}^2$. We first obtain the single vehicle perception results from the N vehicles via different methods in training and inference (specified in subsection III-C). Subsequently, vehicles are dynamically clustered into M groups based on their real-time spatial proximity and the density of the scene. Within each cluster M_i, the most informative vehicle is identified through comparative evaluation of individual perception results. Finally, selected representative features are integrated with the ego vehicle's own features, and the fused information is fed into the detection decoder to produce the final collaborative perception result. An overview of our proposed architecture can be found in Figure 1.

B. Dynamic Clustering and Selection

Our method's core contribution lies in the dynamic clustering and vehicle selection strategy to prune redundant collaborators before fusion. First we form compact clusters that reflect spatial overlap under the current traffic density. Then we elect a single representative in each cluster so that only the most informative senders transmit features to the ego agent.

1) Cluster Formation: cluster formation, we compute the pairwise Euclidean distance between all neighboring vehicles at each time frame,

$$d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2, \tag{1}$$

where $\mathbf{p}_i$ and $\mathbf{p}_j$ denote the spatial positions of vehicles i and j. We next define a density adaptive scaling factor $\alpha(N)$ by:

$$\alpha(N) = \frac{\alpha_0}{1 + \log\bigl(1 + N/N_0\bigr)}, \tag{2}$$

where α_0 is a base scaling constant and N_0 is a normalization parameter tied to typical vehicle density. This choice makes $\alpha(N)$ decrease as density grows and thereby adapts the effective scale to the scene. By selecting α_0 and N_0 appropriately, the sensitivity of clustering can be matched to different environments and bandwidth budgets. The adaptive clustering radius r_c is calculated accordingly as:

$$r_c = \alpha(N) \cdot \frac{1}{N} \sum_{i=1}^{N} \min_{j \neq i}(d_{ij}), \tag{3}$$

where the nearest neighbor term supplies an average baseline spacing. Intuitively, in dense scenes the radius tightens so that only very close vehicles with strongly overlapping views are grouped, which prevents over merging of loosely related neighbors. In sparse scenes $\alpha(N)$ is larger and the radius expands, which avoids over splitting and preserves coverage. Vehicles are then clustered with DBSCAN using $\epsilon = r_c$ and a minimum cluster size of one. In practice this produces clusters that remain stable over time, since both the distance statistics and the adaptive scale evolve smoothly with the scene, and it requires no signals beyond positions and distances.

2) Representative Selection: For each dynamically formed cluster, one vehicle is chosen as the representative according to relative informativeness. During training the representative is defined as the vehicle that covers the largest number of ground truth bounding boxes inside the cluster. This rule leverages supervision to form a consistent prior about which viewpoint contributes the richest cues for the objects present. During inference we replace ground truth counts with an aggregated detection confidence that sums the scores of the vehicle's detections within the shared sensing region. This score summarizes both the reliability of the detector and the amount of potentially useful content observed by that vehicle at the current time. Only the selected representatives transmit intermediate features to the ego and non representatives in the same cluster remain silent. This decoupled selection strategy allows ground truth to establish spatially robust priors

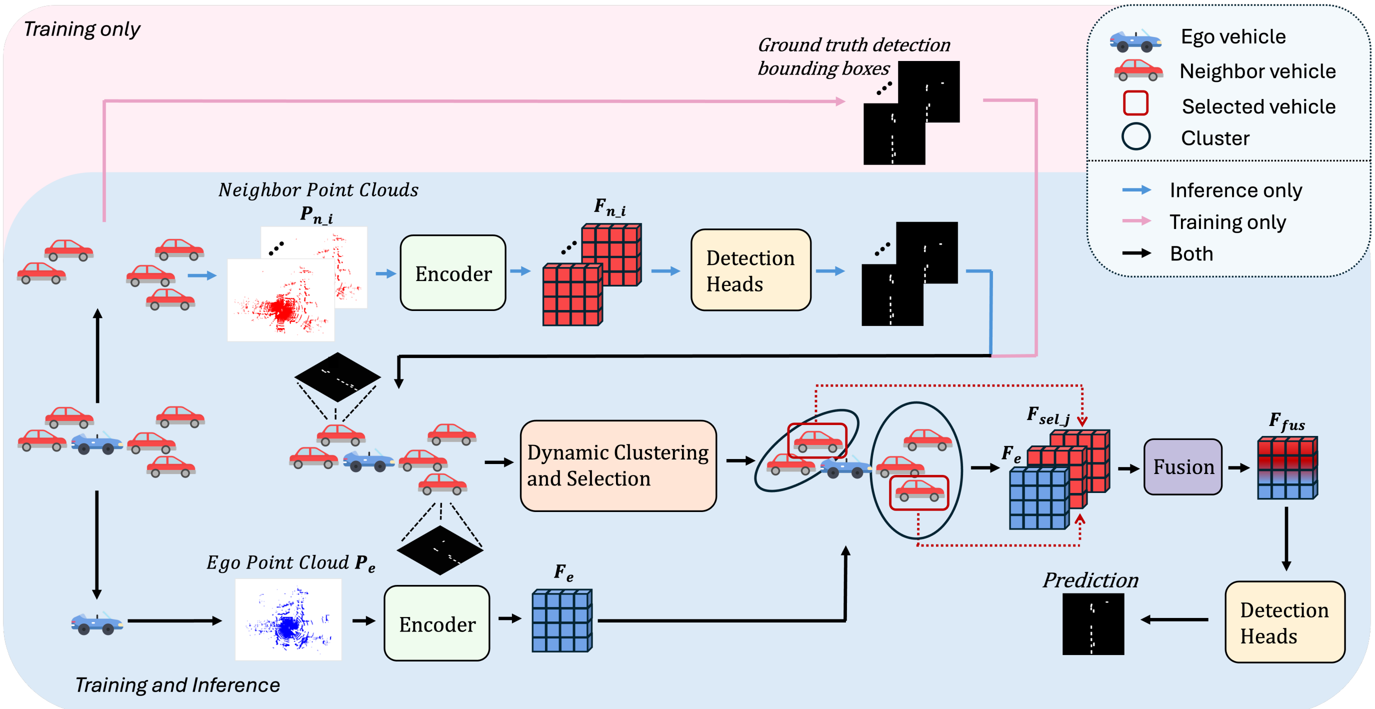

Fig. 1: Overview of our proposed FENSe. During Training, ground truth detection bounding boxes guide dynamic clustering and representative vehicle selection for better learning. During inference, each neighbor independently performs single-vehicle detection and sends compact metadata to the ego vehicle. For both training and inference, the ego vehicle dynamically clusters neighbors, selects one representative per cluster, and requests full features only from the selected vehicles. Selected features are fused and decoded by detection heads to yield the final collaborative detections.

during training, while confidence-based metrics enable real-time adaptation to perceptual uncertainty during deployment. Detailed implementation is provided in Section III-C.

We argue that selecting only one vehicle per cluster is sufficient because vehicles in close spatial proximity typically share highly overlapping fields of view. By appropriately tuning the clustering radius, we ensure that only vehicles with similar viewpoints are grouped together. While this approach may overlook subtle differences in perception due to slight variations in viewpoint within a cluster, it effectively eliminates redundant information without the overhead of comparing full feature maps or engaging in heavy inter-agent communication. This lightweight design significantly reduces the number of features passed to the fusion stage, which often relies on attention mechanisms [27] that are computationally intensive.

C. Training and Inference Time Implementations

1) Training Pipeline: During training we use ground truth annotations to ensure vehicle selection is precise and reproducible. We first form perspective clusters $\mathcal{C}_m$ from ground truth vehicle positions $\mathbf{p}_i^{gt}$ at each frame so that neighbors with similar viewpoints are grouped. Inside each cluster $\mathcal{C}_m$ we measure informativeness by the ground truth bounding box count B_v for each neighboring vehicle v_k, and we choose the representative by maximizing this count

$$v_m^* = \arg\max_{v_k \in \mathcal{C}_m} B_v. \tag{4}$$

This supervised rule anchors the policy on the source that covers the most relevant objects and, for a fixed annotation set, it is deterministic, which stabilizes the stream of features presented to the fusion module during learning. Features $\mathbf{F}_k$ from non representatives $v_k \neq v_m^*$ are filtered at the source so exactly one tensor per cluster is forwarded.

For fusion we adopt attention based per location aggregation in line with prior work [5], [9], [19], [26]. However, these strategies remain sensitive to localization noise and often neglect the inherent sparsity of LiDAR-based point cloud features. Following [5], [8], for each selected neighbor with feature $\mathbf{F}_k^{(t)}$ we construct a confidence map $\mathbf{S}_k^{(t)}$ and a binary selection mask $\mathbf{C}_k^{(t)}$ and remove low confidence elements through

$$\tilde{\mathbf{F}}_k^{(t)} = \mathbf{F}_k^{(t)} \odot \mathbf{C}_k^{(t)}. \tag{5}$$

To guide spatially aware fusion we encode both $\tilde{\mathbf{F}}_k^{(t)}$ and $\mathbf{S}_k^{(t)}$ into multi scale representations $\tilde{\mathbf{F}}_{k,l}^{(t)}$ and $\mathbf{S}_{k,l}^{(t)}$ for $l = 1, 2, 3$ so that fine and coarse structure is exposed without increasing the number of senders. We then form a global confidence by summing over neighbors

$$\mathbf{S}^{(t)}_{\text{sum},l} = \sum_{k=1}^{K} \mathbf{S}^{(t)}_{k,l}, \tag{6}$$

and we select high confidence query locations with the thresholding function f_{sel} following [8]:

$$q \in \mathbf{S}^{(t)}_{\text{loc},l} = f_{\text{sel}}\big(\mathbf{S}^{(t)}_{\text{sum},l}\big). \tag{7}$$

To handle feature misalignment and improve robustness to pose noise, deformable cross attention aggregates information from adaptively sampled offsets around each query [27]. For each reference point q we learn offsets Δq_m that sample from neighboring agents and we assign weights that determine which neighbor and which offset carry the most useful content, producing the fused representation

$$Z^{(t)}_i(q) = \sum_{k=1}^{K}\sum_{m=1}^{M} \mathbf{w}_{k,m}(q)\cdot \tilde{\mathbf{F}}^{(t)}_{k,l}\big(q+\Delta q_m\big), \tag{8}$$

where the weights $\mathbf{w}_{k,m}(q)$ come from a softmax over learned attention scores derived from the projected query features. This concentrates computation on a small set of spatial samples per query rather than on the full grid and aligns well with the sparsity of LiDAR features.

For greater detection precision we incorporate historical features with a recurrent unit in the style of [8]. We then combine the history, the newly fused features, and the ego backbone features by generating spatial importance maps

$$\mathbf{H}^{(t)}_i,\ \mathbf{Z}^{(t)}_i,\ \mathbf{S}^{(t)}_i = \sigma\big(f_{\text{gen}}(\mathbf{H}^{(t)}_i,\ \mathbf{Z}^{(t)}_i,\ \mathbf{F}^{(t)}_i)\big), \tag{9}$$

and after softmax normalization across the three sources we obtain the final fused feature

$$\hat{\mathbf{F}}^{(t)}_i = \mathbf{E}^{(t)}_H \odot \mathbf{H}^{(t)}_i + \mathbf{E}^{(t)}_Z \odot \mathbf{Z}^{(t)}_i + \mathbf{E}^{(t)}_S \odot \mathbf{F}^{(t)}_i, \tag{10}$$

where $\mathbf{E}^{(t)}$ provides the normalization weight for each component. The resulting tensor is forwarded to the classification and regression heads to decode detection outputs.

2) Inference Procedure: At inference time each neighbor executes local single vehicle perception using the same detection backbone that was trained for collaboration, with the fusion module turned off. Specifically, each vehicle first preprocesses its raw LiDAR point cloud to generate voxelized representations. These voxel features are then passed through a feature extraction network to produce high-dimensional feature maps. Subsequently, a region proposal network is applied to generate a set of 3D bounding box proposals. Each proposed bounding box is associated with a detection confidence score to indicate the likelihood of the existence of a valid object.

For filtering and neighbor choice the ego does not require precise localization of every proposal nor the full parameter vector for each box. Instead each neighbor summarizes its scene with a single informativeness value and its pose. The informativeness value is a sum of proposal scores above a fixed threshold and reflects both detector reliability and the amount of potentially useful content observed at that instant.

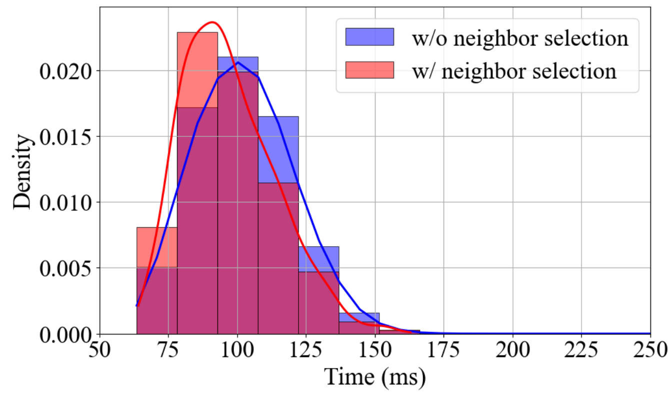

Fig. 2: Computation time at the ego vehicle for processing transmitted neighbor features, with and without our proposed FENSe component. *Blue*: time required to fuse features from all neighbors. *Red*: time for clustering and selection, and fusion of selected neighbor's features.

$$S_i = \sum_{b\in\mathcal{D}_i,\ \sigma(b)>\tau} \sigma(b), \tag{11}$$

where the confidence $\sigma(b)$ is produced by the backbone and the selection threshold τ is fixed on a held out split. Each neighbor transmits the pair formed by its scalar S_i and its position vector $\mathbf{p}_i$. The message is compact and does not include features. Upon receiving messages, inside each cluster, the ego selects the neighbor that has the largest aggregated score S_i as the representative sender for that frame. All other neighbors in the cluster remain silent for that frame in order to save bandwidth and compute. Full perceptual data are only requested from these selected representatives. Finally, features from selected representatives are integrated with the ego vehicle's own perception data using the optimized fusion mechanisms (as described in the training pipeline) to provide collaborative perception outputs.

IV. Experiment

A. Dataset and Evaluation Metrics

We validate the performance of our proposed FENSe on the benchmark dataset OPV2V [9] for collaborative perception tasks. OPV2V is built using the CARLA simulator [28] and the OpenCDA [29] framework, providing a diverse range of urban driving scenarios. The dataset contains a total of 11,464 annotated frames, each equipped with synchronized 64-line LiDAR point clouds and RGB images, along with 3D bounding box annotations for vehicles. The dataset encompasses 73 diverse scenes and involves various number of connected autonomous vehicles (CAVs) per scene, which is suitable for the evaluation of our dynamic neighbor selection framework. We provide results on the testing set of OPV2V.

For evaluation metrics, we use Average Precision (AP) at Intersection-over-Union (IoU) thresholds of 0.3, 0.5 and 0.7 to

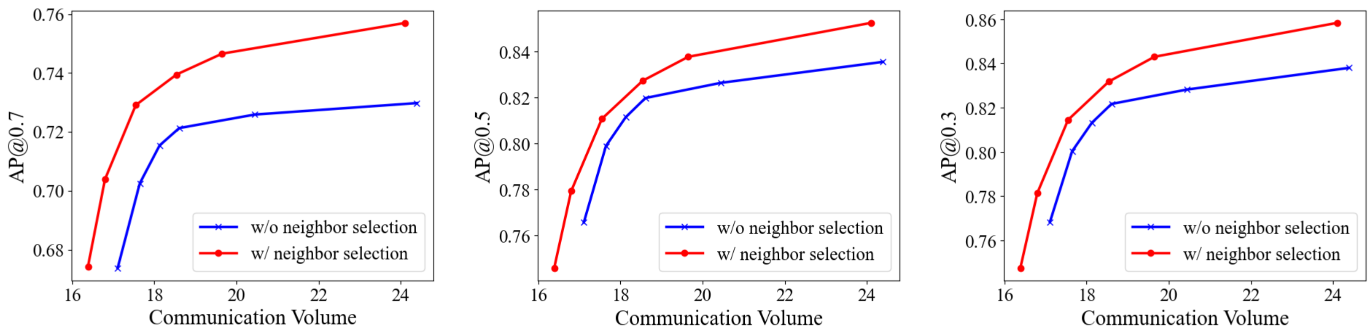

Fig. 3: Collaborative perception performance under different communication volume with and without our proposed FENSe component for dynamic neighbor selection.

measure detection accuracy, time in milliseconds for system efficiency, and communication volume in bytes for evaluation of transmission cost.

B. Implementation Details

Our proposed module is designed to be seamlessly integrated into any collaborative perception framework that follows the standard pipeline of LiDAR point cloud acquisition, feature encoding and selection, information fusion, and detection decoding. In this study, we validate the effectiveness of FENSe within the SCOPE framework [8], which leverages both spatial and temporal information to inform model learning. We adopt the related design and configuration as described in their original implementation.

All experiments are conducted using one NVIDIA GeForce RTX 4090 GPU. For training on the OPV2V dataset [9], we set the batch size to 3 and train the model for 15 epochs. To simulate typical collaboration noise, we model the collaborators' localization and heading errors as independent Gaussian distributions with standard deviations of $0.2\,\mathrm{m}$ and $0.2°$, respectively. For the clustering parameters, we set the base scaling constant α_0 as 2 and N_0 as 3. The bounding box score threshold τ is set to 0.20 for aggregate detection confidence score computing. The spatial positions of the vehicles include only their x and y coordinates to compute pairwise Euclidean distances in the 2D plane.

C. Quantitative Results

Detection accuracy: To demonstrate the performance of FENSe, the following baseline models are designed for comparison:

1) The existing state-of-the-art (SOTA) models [5], [8], [19], [20] without the FENSe module.
2) The [8] model with 30% random dropout of neighbor vehicles.
3) The [8] model with 50% random dropout of neighbor vehicles.
4) Single vehicle perception model without fusion.

Table I presents the 3D detection performance of all evaluated models. Due to SCOPE's [8] superiority of performance, we integrate FENSe on SCOPE for comparison. As expected, our proposed FENSe module substantially improves collaborative perception performance over the single-vehicle baseline, which relies solely on the ego vehicle's spatio-temporal point cloud information. Notably, the integration of our FENSe component on the original SCOPE [8] framework significantly boosts performance by 2.5% in AP@0.3, 2% in AP@0.5, and 3.8% in AP@0.7. The larger gain at higher IoU suggests that FENSe primarily benefits localization precision, which is consistent with the design that concentrates fusion on a single representative per cluster to suppress redundant, slightly misaligned features from overlapping neighbors. To assess the effectiveness of our neighbor selection strategy, we also compare against variants of SCOPE with randomly dropped neighbors. It is observed that higher rates of random dropout lead to greater performance degradation, indicating that naive or inadequate filtering eliminates valuable information in the fusion process. In contrast, integrating FENSe with spatially informed dynamic neighbor selection yields superior performance, demonstrating its effectiveness in preserving the most informative features.

TABLE I: Comparison of AP under Different IoU Thresholds. The incorporation of our module gives the best overall performance across all thresholds.

AP / Models	0.3	0.5	0.7
V2X-ViT [19]	82.64	80.93	67.79
CoBEVT [20]	82.96	81.60	71.37
Where2Comm [5]	83.03	82.12	69.29
SCOPE [8]	83.80	83.54	72.98
SCOPE+Random 30% Neighbor Drop	83.46	83.24	70.25
SCOPE+Random 50% Neighbor Drop	81.66	80.93	69.38
No Fusion	76.19	74.21	60.50
SCOPE + FENSe (ours)	**85.83**	**85.23**	**75.70**

Computation Time at the Ego Vehicle: Efficient processing at the ego vehicle is essential for ensuring real-time performance in collaborative perception systems. While the integration of FENSe reduces computation time by reducing the volume of feature data involved in the fusion process, it also introduces additional overhead due to clustering and neighbor selection. To assess the overall impact of our module on collaborative perception latency, we evaluate the processing time at the ego vehicle. In our framework, we assume that each vehicle completes its single-agent perception prior to participating in collaboration. Therefore, we measure computation time at the ego vehicle starting from the moment it receives features transmitted by its neighbors, and ending when the final fused feature map is produced.

Figure 2 presents a histogram of the ego-side processing time across all frames and scenes in the test set, where the number of collaborating neighbors varies from scene to scene. Although the integration of our FENSe module introduces additional operations for clustering and neighbor selection, it decreases the overall processing time. This reduction in runtime is attributed to the fact that FENSe significantly reduces the number of features that need to be processed at the ego side by selecting only a subset of informative neighbors. Therefore, the overhead introduced by clustering and selection is outweighed by the savings in fusion computation.

Communication Volume: Communication volume is an important constraint in collaborative perception settings. To evaluate the transmission cost of our proposed FENSe module, we follow [5] to compute the message size in bytes in log scale with base 2. Specifically, we adjust the threshold of the binary selection matrix $\mathbf{C}_{\mathbf{k}}^{(\mathbf{t})}$ of selected neighbor vehicles to emulate varying bandwidth constraints. As shown in Figure 3, FENSe consistently achieves higher detection performance at given communication volumes across all IoU thresholds: for a fixed communication budget, AP is consistently higher; conversely, to reach a target AP, FENSe requires fewer bytes. The gap widens at larger budgets because the baseline increasingly transmits overlapping features from multiple neighbors, whereas FENSe maintains sender-level sparsity, preventing redundant payloads from ever entering the channel. Overall, when compared to the baseline without neighbor selection, FENSe simultaneously improves AP and reduces communication, demonstrating that eliminating inter-agent redundancy before feature exchange is a more effective lever than per-sender, message-level filtering alone.

TABLE II: Ablation study results of the proposed components or procedures.

Models \ AP	0.3	0.5	0.7
w/o training selection	82.74	82.43	72.01
w/o inference selection	85.88	85.40	75.66
w/o dynamic scaling factor	86.27	85.00	72.24
ours	**85.83**	**85.23**	**75.70**

Ablation Studies: We conduct an ablation study to assess the contribution of various components and stages of our dynamic neighbor selection module. The results are summarized in Table II.

Removing training-time selection (*w/o training selection*) leads to a notable drop in performance across all IoU thresholds, especially at higher precision levels. This confirms that supervising the model with cluster representatives during training teaches it to depend on representative, non-redundant collaborators and reduces over-reliance on overlapping senders. Mechanistically, without this supervision the fusion module sees many near-duplicate neighbor features, which increases attention competition and gradient noise and ultimately harms precise localization

Disabling selection only at inference (*w/o inference selection*) admits *all* neighbors to fusion. Interestingly, this causes a slight increase in perception at low and medium IoU thresholds: AP@0.3 increases relatively by approximately 0.06%, AP@0.5 increases by 0.20%, whereas AP@0.7 decreases by 0.05%. The small bumps at low and medium IoU arise because enabling communication with all neighbors provide richer information and the extra features occasionally recover marginal detections. However, this gain comes at the cost of significantly increased communication overhead and computational burden at the ego vehicle. In practical applications with limited bandwidth, it is often preferable to trade off a small amount of precision for substantial savings in communication and processing time.

Removing the dynamic scaling factor (*w/o dynamic scaling factor*) results in a sharp performance drop at higher IoU thresholds. A non-adaptive radius under-merges in dense scenes because too many representatives are admitted, increasing coarse recall but injecting redundant and slightly misaligned features that hurt precise box alignment. Conversely, it can over-merge in sparse scenes and miss complementary viewpoints. This indicates that adaptively controlling the clustering scale is particularly important for fine-grained perception.

V. Conclusion and Future Work

In this work, we presented *FENSe*, a lightweight feedback-enabled dynamic neighbor selection module that removes inter-agent redundancy before feature exchange. By turning single-vehicle perception into compact feedback and pairing it with density-aware clustering, the ego vehicle selects one representative per spatial cluster and requests full features only from those representatives. The design is simple to implement, integrates seamlessly with standard collaborative perception pipelines, and does not require additional feature-level operations on high-dimensional maps. Through extensive experiments, we demonstrated that FENSe consistently improves detection performance while reducing communication volume and computation time at the ego vehicle. FENSe offers a practical control point for who should communicate in multi-agent collaborative perception and can complement existing content-selection policies.

Currently, FENSe forms spatial clusters with a density aware radius and then chooses one representative within each cluster. While this delivers a clear reduction in duplicated features, a side effect is that two neighbors with overlapping viewpoints but different occlusions can be grouped together, which may conceal a small amount of complementary evidence due to neighbor selection. The adaptive radius mitigates this effect by tightening in dense traffic and expanding in sparse scenes, and the feedback loop can change the representative as conditions evolve. In future works, temporal continuity can be introduced into selection so that decisions reflect how observations evolve over consecutive frames and informative evidence can be preserved.

Additionally, FENSe also inherits a security exposure that is common in cooperative systems. It assumes honest metadata from collaborators. A malicious or poisoned neighbor can inflate confidence or fabricate coverage to become the representative and then transmit misleading features. Future work can address this by incorporating reliability assessment grounded in consistency over time and cross checking with recent context, so that manipulated signals are less likely to influence the chosen representative. With learned temporal regularities, the system can judge unusual outliers and additional misses more confidently and can revisit earlier choices when new evidence appears.

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